

Assignment 2

1. Show that every continuous homomorphism from $\mathbb{R}$ to itself is of the form $x \mapsto \alpha x$ for some fixed $\alpha \in \mathbb{R}$
2. Show that any finite subgroup of $\mathbb{C}^*$ is cyclic.
3. Let G be an abelian group which has elements of order m, n . Show that G has an element of order $\text{lcm}(m, n)$.
4. In the group $G = \mathbb{C}^*$, find the cosets of the subgroup $H = \{z \in G \text{ such that } |z| = 1\}$ and describe them geometrically.
5. Let H be a subgroup of G such that $x^2 \in H$ for all $x \in G$. Show that H is a normal subgroup and G/H is abelian.
6. Let G be a finite group. Suppose for all $x \in G$, there exists $y \in G$ such that $y^2 = x$, then G has odd order, and conversely.